

Task-1

Step-by-Step Guide: Blocking Spotify Traffic

1. Open Windows Defender Firewall with Advanced Security:

- Press Win + R on your keyboard to open the **Run** dialog.
- Type wf.msc and press **Enter** (or click **OK**).

2. Start the New Outbound Rule Wizard:

- In the left-hand pane, click on **Outbound Rules**.
- In the right-hand **Actions** pane, click **New Rule...**

3. Configure the Rule Settings:

- **Rule Type:** Select **Program** and click **Next**.
- **Program Path:** Select **This program path:**, click **Browse...**, and navigate to the Spotify executable file:
 - *Default* *location:*
C:\Users\<YourUsername>\AppData\Roaming\Spotify\Spotify.exe
 - *(Tip: If installed via Microsoft Store, the executable path is inside C:\Program Files\WindowsApps).*
 - Click **Next**.
- **Action:** Select **Block the connection** and click **Next**.
- **Profile:** Ensure **Domain**, **Private**, and **Public** are all checked, then click **Next**.
- **Name:** Type a clear name like Block Spotify Outbound and click **Finish**.

Verification

Once the rule is active, open the **Spotify** desktop app. The app will fail to stream audio, connect to servers, or fetch online content, confirming that the outbound rule is actively blocking its network traffic.

TASK-2

Note: Windows Defender Firewall rules cannot reliably restrict traffic to **only Google Chrome on port 443** using the basic GUI because HTTPS (port 443) is used by many applications. However, you can create a rule that allows **Google Chrome** to receive inbound connections. If you specifically want to restrict it to **TCP port 443**, you can create a custom rule.

Steps to Create an Inbound Rule for Google Chrome on Port 443

1. Press **Win + R**, type **wf.msc**, and press **Enter** to open **Windows Defender Firewall with Advanced Security**.
2. In the left pane, click **Inbound Rules**.
3. In the right pane, click **New Rule....**
4. Select **Custom**, then click **Next**.
5. Under **Program**, choose **This program path**, then browse to the Chrome executable:
6. C:\Program Files\Google\Chrome\Application\chrome.exe
(or C:\Program Files (x86)\Google\Chrome\Application\chrome.exe on some systems).
7. Click **Next**.
8. Under **Protocol and Ports**:
 - Protocol type: **TCP**
 - Local Port: **Specific Ports → 443**
9. Click **Next**.
10. Under **Action**, select **Allow the connection**.
11. Click **Next**.
12. Select the profiles where the rule should apply (**Domain**, **Private**, and/or **Public**).
13. Click **Next**.

14. Give the rule a name, such as:
15. Allow Google Chrome HTTPS (Inbound)
16. Click **Finish**.

Result

A new inbound firewall rule is created that allows **Google Chrome** to receive **TCP traffic on port 443 (HTTPS)** for the selected network profiles.

Note for assignments: In normal web browsing, **Google Chrome usually initiates outbound HTTPS connections**, so creating an **outbound rule** is generally more practical than an inbound rule. However, the above steps satisfy the requested task of creating an inbound rule.

TASK-3

Blocking Traffic from a Specific IP Address Using Windows Defender Firewall

Example IP Address: 123.45.67.89

Step 1: Open Windows Defender Firewall

1. Press **Win + R**.
2. Type **wf.msc** and press **Enter**.
3. The **Windows Defender Firewall with Advanced Security** window will open.

Step 2: Create a New Rule

1. Click **Inbound Rules** in the left pane.
2. In the right pane, click **New Rule....**

Step 3: Select Rule Type

1. Choose **Custom**.
2. Click **Next**.

Step 4: Select the Program

1. Choose **All Programs** (or select a specific program if needed).
2. Click **Next**.

Step 5: Configure Protocol and Ports

1. Leave **Protocol Type** as **Any**.
2. Leave **Local Port** and **Remote Port** as **Any**.
3. Click **Next**.

Step 6: Specify the IP Address (Scope)

1. Open the **Scope** tab.
2. Under **Which remote IP addresses does this rule apply to?**
3. Select **These IP addresses**.
4. Click **Add**.
5. Enter the IP address:
6. 123.45.67.89
7. Click **OK**, then **Next**.

Step 7: Choose the Action

1. Select **Block the connection**.
2. Click **Next**.

Step 8: Select Network Profiles

Choose the profiles where the rule should apply:

- ☒ Domain
- ☒ Private
- ☒ Public

Click **Next**.

Step 9: Name the Rule

Enter a descriptive name, for example:

Block IP 123.45.67.89

Optionally add a description, then click **Finish**.

Verification

- The new firewall rule will appear under **Inbound Rules**.
- Any incoming traffic from **123.45.67.89** will now be blocked.
- To block **outgoing** traffic to the same IP address, repeat the same steps under **Outbound Rules**.

TASK-4

Hardware Firewall vs Software Firewall

Feature	Hardware Firewall	Software Firewall
Location	A dedicated physical device installed between the Internet and the network.	A program installed on an individual computer or laptop.
Protection	Protects all devices connected to the network.	Protects only the device on which it is installed.
Performance & Management	Handles large amounts of network traffic and is centrally managed.	Uses the computer's resources and is managed separately on each device.

Use Cases

Firewall Type	Use Case
Hardware Firewall	Used in large offices, schools, and businesses to protect the entire network from unauthorized access and cyber threats.
Software Firewall	Used on home computers and laptops to block unwanted connections and protect the individual device while browsing the internet or using applications.

Summary

- A **hardware firewall** is best for protecting an **entire network** with centralized security.
- A **software firewall** is best for protecting **individual computers** and is commonly included with operating systems such as **Windows Defender Firewall**.